ABSTRACT OF THE DISCLOSURE

A method and system for establishing a high-bandwidth, short-range, free-space optical data link between two uncoupled mobile electronic devices, such as smartphones or tablets. The system utilizes the screen of a transmitting device as a frame-based optical emitter and the camera of a receiving device as an optical sensor. Data is encoded into a sequence of frames displayed on the transmitter's screen, where each frame represents a large block of parallel data. To ensure reliability in an unguided, variable environment, the system uses large "channel elements"—grouping blocks of physical pixels into logical super-pixels—to compensate for movement, distance, and angular misalignment. A negotiation and synchronization protocol is established using a preliminary handshake (e.g., displaying a QR code) to allow the receiver to lock on to the transmitter's display and calibrate for the data transfer, enabling multi-gigabit-per-second transfer rates for tasks like migrating the entire contents of a device.